Unit 2 Quiz A

Forces & Translational Dynamics · AP-style multiple choice

12 questions + 1 optional Time: **24 min** (AP pace ≈ 2 min each) Calculator OK g = 9.8 m/s²

- Before choosing, **write one line of work** in the margin: an FBD sketch, an equation, or a ratio. Those are the habits the FRQs grade.
- Circle your confidence for every question. A lucky guess still counts as a gap.
- Grade with the coaching guide. For every miss AND every 'guess', read the full walkthrough and do the 'Go deeper' question.

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1. `2.1 Center of mass`

Three small objects sit on a light rod: 2.0 kg at x = 0, 3.0 kg at x = 2.0 m, and 5.0 kg at x = 4.0 m. Where is the center of mass of the system?

A x = 2.0 m **B** x = 2.6 m **C** x = 3.25 m **D** x = 4.0 m

Confidence: sure | 50/50 | guess Answer: ____

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2. `2.2 / 2.4 FBD + equilibrium`

A crate is dragged across a rough floor at CONSTANT velocity by a rope that pulls with tension T at angle θ above the horizontal. Which statement is correct?

A F_N = mg and the friction force equals T

B F_N < mg and the friction force equals T cos θ

C F_N > mg and the friction force equals T cos θ

D F_N < mg and the friction force is less than T cos θ

Confidence: sure | 50/50 | guess Answer: ____

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3. `2.3 Newton's 3rd law`

Alex (50 kg) and Blake (100 kg) sit on rolling office chairs facing each other. Alex pushes on Blake's shoulders and they roll apart. While the hands are in contact, which is true?

A Alex pushes harder on Blake than Blake pushes on Alex, and they have equal accelerations

B The forces are equal in size, and Alex's acceleration is twice Blake's

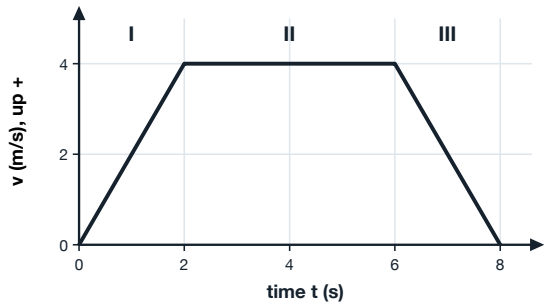
C The forces are equal in size, and their accelerations are equal

D Blake pushes harder, because Blake is heavier, and Alex's acceleration is larger

Confidence: sure | 50/50 | guess Answer: ____

4. `2.5 Newton's 2nd law + graphs`

The graph shows the velocity of an elevator (up is positive). A rider stands on a scale. Rank the scale reading during intervals I, II and III.



A I > II > III **B** I = II = III **C** III > II > I **D** II > I > III

Confidence: sure 50/50 guess Answer: ____

5. `2.6 Gravitational field`

Planet X has twice Earth's mass and twice Earth's radius. What is the gravitational field strength at its surface? (Earth: g = 9.8 N/kg)

A 19.6 N/kg **B** 9.8 N/kg **C** 4.9 N/kg **D** 2.45 N/kg

Confidence: sure 50/50 guess Answer: ____

6. `2.6 Apparent weight`

Astronauts float inside the International Space Station, about 400 km above Earth's surface (Earth's radius is about 6400 km). Which explanation is correct?

A There is no gravity in space

B Gravity at that height is only about 10% of its value at the surface

C The astronauts and the station are both in free fall at the same acceleration, so nothing pushes on the astronauts

D The centripetal force on the astronauts cancels their weight

Confidence: sure 50/50 guess Answer: ____

7. `2.6 Universal gravitation`

A satellite weighs W on Earth's surface. It is launched to an altitude ABOVE THE SURFACE equal to one Earth radius. What is the gravitational force on it there?

A W **B** W / 2 **C** W / 4 **D** 0

Confidence: sure 50/50 guess Answer: ____

8. `2.7 Static friction`

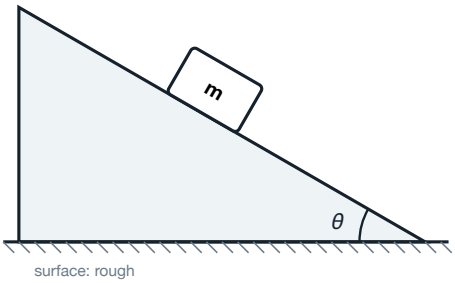
A 5.0 kg box rests on a floor ($\mu_s = 0.50$, $\mu_k = 0.30$). You push it horizontally with 20 N and it does not move. What is the friction force on the box?

A 0 N **B** 14.7 N **C** 20 N **D** 24.5 N

Confidence: sure 50/50 guess Answer: ____

9. `2.7 Friction on an incline`

A block of mass m rests, not moving, on a rough ramp tilted at angle θ. The coefficient of static friction is μ_s. What is the magnitude of the friction force on the block?

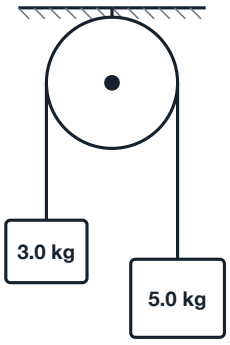


A μ_s mg cos θ **B** mg sin θ **C** mg cos θ **D** μ_s mg sin θ

Confidence: sure 50/50 guess Answer: ____

10. `2.5 Connected systems`

A 3.0 kg block and a 5.0 kg block hang from opposite ends of a light string over an ideal pulley and are released. What is the tension in the string?

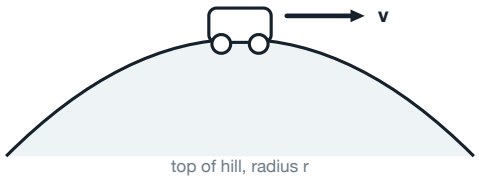


A 29.4 N **B** 36.8 N **C** 39.2 N **D** 49.0 N

Confidence: sure 50/50 guess Answer: ____

11. 2.9 Circular motion

A car of mass m drives over the top of a rounded hill of radius r at speed v. What is the normal force from the road at the very top?

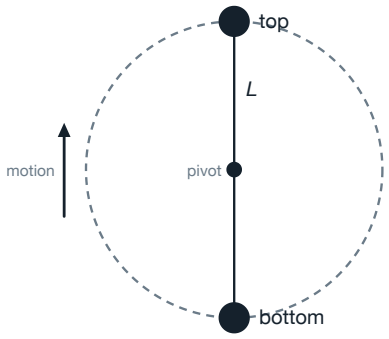


A mg **B** mg + mv²/r **C** mg – mv²/r **D** mv²/r

Confidence: sure 50/50 guess Answer: ____

12. 2.9 Vertical circle · BOSS

A ball on a string moves in a VERTICAL circle at constant speed. Rank the string tension at the top (T_{top}), at the side where the string is horizontal (T_{side}), and at the bottom (T_{bot}).



A $T_{top} > T_{side} > T_{bot}$ **B** $T_{bot} > T_{side} > T_{top}$

C All equal, because the speed is constant **D** $T_{bot} > T_{top} > T_{side}$

Confidence: sure 50/50 guess Answer: ____

13. 2.8 Springs · optional: ask if springs are on the test

A 0.50 kg mass hangs at rest from a spring and stretches it 0.10 m. What is the spring constant, and how far would a 1.5 kg mass stretch the same spring?

A 49 N/m; 0.30 m **B** 5.0 N/m; 0.30 m

C 49 N/m; 0.90 m **D** 4.9 N/m; 0.15 m

Confidence: sure 50/50 guess Answer: ____